

WASP-2b

R-0124

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_250711 · created 2026-07-16T05:40:23+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460867.8546 +/- 0.0074 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.16 +/- 0.043
Transit depth (Rp/Rs)^2	2.56 +/- 1.38 [%]
Orbital Inclination (inc)	83.98 +/- 2.47
Airmass coefficient 1 (a1)	0.942 +/- 0.021
Airmass coefficient 2 (a2)	0.05 +/- 0.1
Scatter in the residuals of the lightcurve fit is	2.25 %
Best Comparison Star	#1 - [288, 404]
Optimal Method	PSF photometry
Transit Duration (day)	0.058 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.16 ± 0.043 vs 0.1326 (deviation 0.61σ)
Duration fit vs published	0.058 d vs 0.0737 d (deviation 0.44σ)
Mid-time O–C	-14.4 min
Depth SNR	3.6
Residual scatter	2.25 %

Light curve

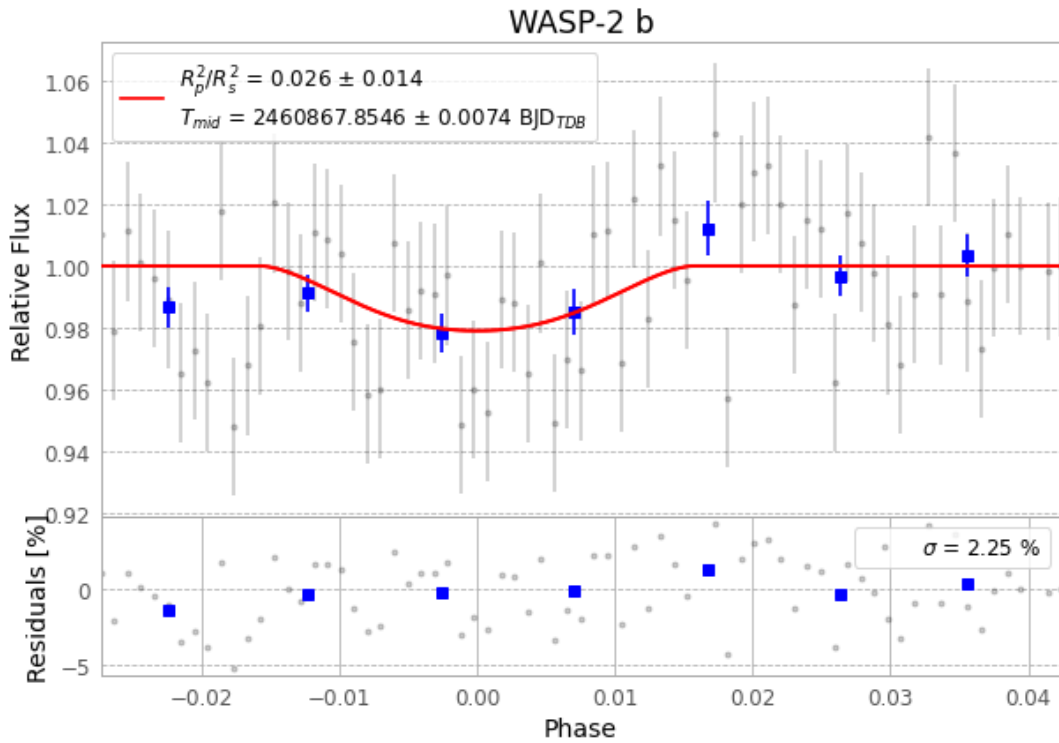


Figure 1 — FinalLightCurve_WASP-2 b_2025-07-10.png (SHA-256 23ff9e2dab8e5a61...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [287, 205], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8e670c21d5284009...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 3ebc8ba

Data provenance

73 FITS frames (48 MB), WASP-2250711065719.FITS → WASP-2250711103315.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-250711.log (142367852ac7...), FinalLightCurve_WASP-2 b_2025-07-10.png (23ff9e2dab8e...), FinalLightCurve_WASP-2 b_2025-07-10.csv (c6833f2371c8...)

Compute resources

Wall time 130.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 05:40Z.